accomplishment is accompanied by the activation of the "vld" indicator, which is demonstrated in lines 136–143.

## PBL 22: Design Simulation on Single-Port Register File

1) Create a testbench to simulate the single-port register file introduced in Section 9.2.2.
2) Create the test case illustrated in Figure 9.5(a), which involves writing four words into the register file followed by reading them out. The write enable signal "we=4'hf" ensures that all four bytes in each data burst are enabled.
3) Create the test case depicted in Figure 9.5(b), which entails writing two words into the register file with "we=4'hf", followed by reading them out. Additionally, conduct writing two words into the register file with "we=4'h3", followed by reading them out.

## PBL 23: Design Simulation on Dual-Port Register File

1) Create a testbench to simulate the dual-port register file introduced in Section 9.2.3.
2) Create the test case illustrated in Figure 9.7, which involves writing four words into the register file followed by reading them out. The write enable signal "wea=4'hf" ensures that all four bytes in each data burst are enabled. It's essential to emphasize that the write and read operations need to overlap to illustrate the concurrent operations between write and read accesses for the dual-port register file.

## PBL 24: FP Matrix-Matrix Subtractor

1) Design a FP matrix-matrix subtractor utilizing the signal-clock-cycle FP operators, denoted as $z = x - y$, where $x$, $y$, and $z$ are $4 \times 4$ FP matrices. The FP subtractor can be used as the design IP. Three single-port memories can be used to store the input matrices

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$x$ and $y$ and the final output matrix $z$.

$$\begin{pmatrix} z_{00} & z_{01} & z_{02} & z_{03} \\ z_{10} & z_{11} & z_{12} & z_{13} \\ z_{20} & z_{21} & z_{22} & z_{23} \\ z_{30} & z_{31} & z_{32} & z_{33} \end{pmatrix} = \begin{pmatrix} x_{00} & x_{01} & x_{02} & x_{03} \\ x_{10} & x_{11} & x_{12} & x_{13} \\ x_{20} & x_{21} & x_{22} & x_{23} \\ x_{30} & x_{31} & x_{32} & x_{33} \end{pmatrix} - \begin{pmatrix} y_{00} & y_{01} & y_{02} & y_{03} \\ y_{10} & y_{11} & y_{12} & y_{13} \\ y_{20} & y_{21} & y_{22} & y_{23} \\ y_{30} & y_{31} & y_{32} & y_{33} \end{pmatrix} \tag{9.4}$$

$$= \begin{pmatrix} x_{00} - y_{00} & x_{01} - y_{01} & x_{02} - y_{02} & x_{03} - y_{03} \\ x_{10} - y_{10} & x_{11} - y_{11} & x_{12} - y_{12} & x_{13} - y_{13} \\ x_{20} - y_{20} & x_{21} - y_{21} & x_{22} - y_{22} & x_{23} - y_{23} \\ x_{30} - y_{30} & x_{31} - y_{31} & x_{32} - y_{32} & x_{33} - y_{33} \end{pmatrix}$$

2) Create a testbench to simulate the designed circuit and verify its functionalities using the provided direct test case below. Represent the FP data using the IEEE-754 format depicted in Table 8.2. Store the columns of each matrix in memory in a row-wise manner.

$$\begin{pmatrix} x_{00} & x_{01} & x_{02} & x_{03} \\ x_{10} & x_{11} & x_{12} & x_{13} \\ x_{20} & x_{21} & x_{22} & x_{23} \\ x_{30} & x_{31} & x_{32} & x_{33} \end{pmatrix} = \begin{pmatrix} 1.0 & 2.0 & 3.0 & 4.0 \\ 1.0 & 2.0 & 3.0 & 4.0 \\ 1.0 & 2.0 & 3.0 & 4.0 \\ 1.0 & 2.0 & 3.0 & 4.0 \end{pmatrix} \tag{9.5}$$

$$\begin{pmatrix} y_{00} & y_{01} & y_{02} & y_{03} \\ y_{10} & y_{11} & y_{12} & y_{13} \\ y_{20} & y_{21} & y_{22} & y_{23} \\ y_{30} & y_{31} & y_{32} & y_{33} \end{pmatrix} = \begin{pmatrix} 0.0 & 1.0 & 2.0 & 3.0 \\ 0.0 & 1.0 & 2.0 & 3.0 \\ 0.0 & 1.0 & 2.0 & 3.0 \\ 0.0 & 1.0 & 2.0 & 3.0 \end{pmatrix} \tag{9.6}$$

## PBL 25: FP Multiplication-Addition Circuit

1) Design a FP multiplication-addition circuit (MAC) utilizing the signal-clock-cycle FP operators, denoted as $z = x^T \times y$, where $x$ and $y$ are FP vectors and $z$ is a FP number. The FP multiplier and adder can be used as design IPs.

$$z = \begin{pmatrix} x0 & x1 \end{pmatrix} \times \begin{pmatrix} y0 \\ y1 \end{pmatrix} \tag{9.7}$$
$$= x0 \cdot y0 + x1 \cdot y1$$

2) Create a testbench to simulate the designed circuit and verify its functionalities using the provided direct test case below. Represent the FP data using the IEEE-754 format depicted in Table 8.2.

$$\begin{pmatrix} x0 \\ x1 \end{pmatrix} = \begin{pmatrix} 1.0 \\ 2.0 \end{pmatrix} \tag{9.8}$$

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$$\begin{pmatrix} y0 \\ y1 \end{pmatrix} = \begin{pmatrix} 3.0 \\ 4.0 \end{pmatrix} \tag{9.9}$$

## PBL 26: FP Matrix-Matrix Multiplier

1) Design a FP matrix-matrix multiplier utilizing the signal-clock-cycle FP operators, denoted as $z = x \times y$, where $x$, $y$, and $z$ are $2 \times 2$ FP matrices. The multiplication-addition circuit in PBL 25 can be used as the design IP.

$$\begin{pmatrix} z_{00} & z_{01} \\ z_{10} & z_{11} \end{pmatrix} = \begin{pmatrix} x_{00} & x_{01} \\ x_{10} & x_{11} \end{pmatrix} \times \begin{pmatrix} y_{00} & y_{01} \\ y_{10} & y_{11} \end{pmatrix} \tag{9.10}$$
$$= \begin{pmatrix} MAC(x_{00} \sim x_{01}, y_{00} \sim y_{10}) & MAC(x_{00} \sim x_{01}, y_{01} \sim y_{11}) \\ MAC(x_{10} \sim x_{11}, y_{00} \sim y_{10}) & MAC(x_{10} \sim x_{11}, y_{01} \sim y_{11}) \end{pmatrix}$$

2) Create a testbench to simulate the designed circuit and verify its functionalities using the provided direct test case below. Represent the FP data using the IEEE-754 format depicted in Table 8.2.

$$\begin{pmatrix} x_{00} & x_{01} \\ x_{10} & x_{11} \end{pmatrix} = \begin{pmatrix} 1.0 & 3.0 \\ 2.0 & 4.0 \end{pmatrix} \tag{9.11}$$

$$\begin{pmatrix} y_{00} & y_{01} \\ y_{10} & y_{11} \end{pmatrix} = \begin{pmatrix} 0.0 & 2.0 \\ 1.0 & 3.0 \end{pmatrix} \tag{9.12}$$

## PBL 27: Power of FP Matrix

1) Design a circuit to compute the power of an FP matrix utilizing the signal-clock-cycle FP operators, denoted as $y = x^2$, where $x$ and $y$ are $2 \times 2$ FP matrices. The FP multiplication-addition circuit in PBL 25 can be used as the design IP.

$$\begin{pmatrix} y_{00} & y_{01} \\ y_{10} & y_{11} \end{pmatrix} = \begin{pmatrix} x_{00} & x_{01} \\ x_{10} & x_{11} \end{pmatrix} * \begin{pmatrix} x_{00} & x_{01} \\ x_{10} & x_{11} \end{pmatrix} \tag{9.13}$$

$$= \begin{pmatrix} MAC(x_{00} \sim x_{01}, x_{00} \sim x_{10}) & MAC(x_{00} \sim x_{01}, x_{01} \sim x_{11}) \\ MAC(x_{10} \sim x_{11}, x_{00} \sim x_{10}) & MAC(x_{10} \sim x_{11}, x_{01} \sim x_{11}) \end{pmatrix}$$

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2) Create a testbench to simulate the designed circuit and verify its functionalities using the provided direct test case below. Represent the FP data using the IEEE-754 format depicted in Table 8.2.

$$\begin{pmatrix} x_{00} & x_{01} \\ x_{10} & x_{11} \end{pmatrix} = \begin{pmatrix} 1.0 & 3.0 \\ 2.0 & 4.0 \end{pmatrix} \tag{9.14}$$

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